

TrES-3b

R-2609

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_260628 · created 2026-08-28T01:19:46+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2461219.725 +/- 0.002 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.162 +/- 0.057
Transit depth (Rp/Rs)^2	2.62 +/- 1.86 [%]
Orbital Inclination (inc)	85.84 +/- 2.86
Airmass coefficient 1 (a1)	1.57804 +/- 0.00011
Airmass coefficient 2 (a2)	-0.4 +/- 0.22
Scatter in the residuals of the lightcurve fit is	0.0023 %
Best Comparison Star	#1 - [395, 442]
Optimal Aperture	1.34
Optimal Annulus	3.0
Transit Duration (day)	0.065 +/- 0.011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.162 $\pm$ 0.057 vs 0.16309 (deviation 0.02 σ)
Duration fit vs published	0.065 d vs 0.0567 d (deviation 0.76 σ)
Mid-time O-C	0.2 min
Depth SNR	2.8
Residual scatter	0.0023 %

Light curve

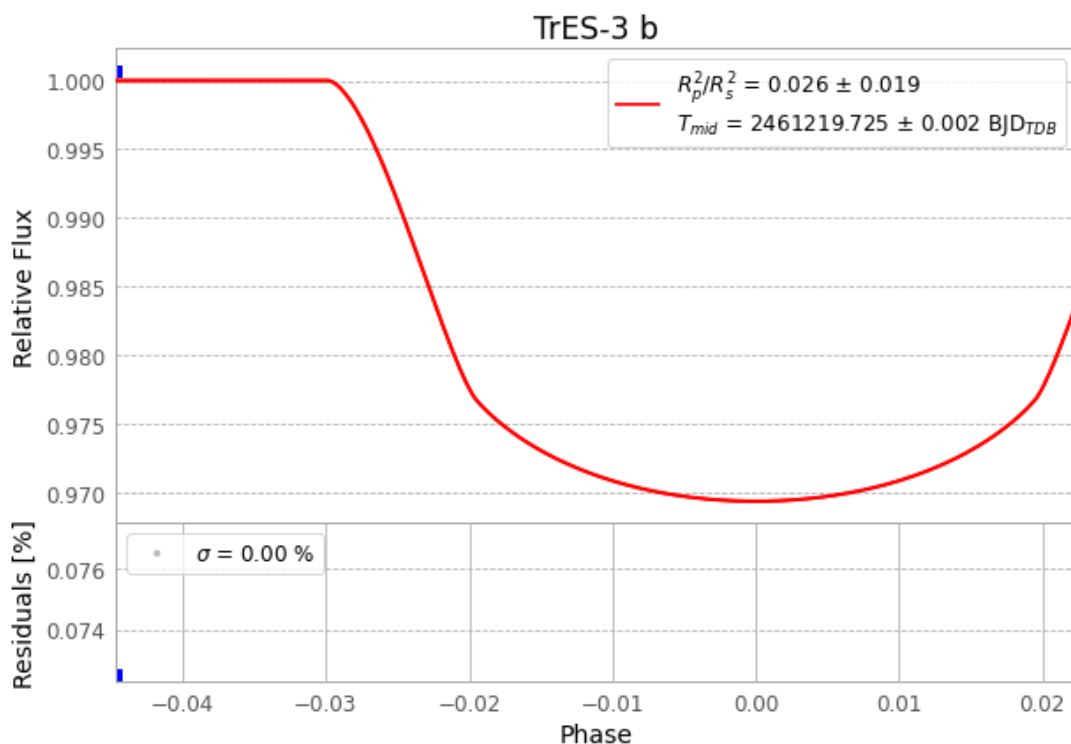


Figure 1 — FinalLightCurve_TrES-3 b_2026-06-27.png (SHA-256 f0cf70091816bb5f...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [288, 284], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 9a5cc53659fbb3d8...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

120 FITS frames (49 MB), TRES-3260628033421.FITS.gz → TRES-3260628070015.FITS.gz. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-260628.log (bb62792fec46...), FinalLightCurve_TrES-3 b_2026-06-27.png (f0cf70091816...), FinalLightCurve_TrES-3 b_2026-06-27.csv (2dd18c5530af...)

Compute resources

Wall time 118.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-28 01:19Z.